INDIAN INSTITUTE OF INFORMATION TECHNOLOGY SONEPAT

भारतीय सूचना प्रौद्योगिकी संस्थान सोनीपत

Email: sonepatiiit@gmail.com, website: www.iiitsonepat.ac.in

Roll no ...12211008

Mid Sem-I Subject: Object Oriented Programming (CSC304)

Branch: CSE 3rd Sem M.M.-15 marks Time- 60 mint.

Note: All questions are compulsory.

Q. 1 Explain the visibility of base class members for the access specifiers: private, protected and public while creating the derived class and also explain the syntax for creating derived class. **3**

Q. 2 List the characteristics of a Constructor and Destructor function. Write a program to define a suitable parameterized constructor with default values for the class Time with data members HH, MM an SS (for Hours, Minutes and Seconds) and try to add two times by passing the arguments (2 objects of time class). **5**

Q. 3 Write a program to display the output as follows **4**

```
      N lines
  ---------------
  --------------
  -------------
  10101010101
   101010101
    1010101
     10101
      101
       1
```

Q.4 What do you mean by function overloading explain it with a suitable example. **3**

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Mid Sem-I **Subject: COMMUNICATION SKILLS (HUL 204)**
Branch: CSE **.M.-15 marks** **Time- 60 min.**
iester II___2022 Batch

Attempt All. **(3 X 5 = 15)**

1. What do you understand by 'perception and visual communication'? Discuss Gestalt and Semiotic theories of visual communication.

2. Write a detailed essay on American English and British English. Elaborate on the differences in the pronunciation of both variants of English.

3. What do you understand by 'barriers to communication? With the help of suitable examples, discuss its various types.

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Mid Sem-I	**Subject Name: WD**	**Subject code: CSC205**	**Date:30 May,2023**
Branch: CSE(2ND SEM)	**M.M.-15 marks**	**Roll no** ..12211008..........	**Time- 60 mint.**

Note: All questions are compulsory.

Q-1:	How XML validation process works? Explain with example.	**(4marks)**
Q-2:	What is HTML form? What are major attribute of form? Explain five components with example.	**(4marks)**
Q-3:	What is CSS? Explain the order in which we can apply style sheet in different ways.	**(2 marks)**
Q-4:	Give the difference between <datalist > tag and <select> tag? Discuss with example	**(2.5marks)**
Q-5:	Write a program to list your favourite games using different styles of unordered list.	**(2.5 marks)**

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MID SEM-1

Branch: CSE 2nd SEM **Subject Name: Data Structure** **Subject Code: CSC203**

M.M.-15 marks **Roll no** 6211006 **Time- 01HRS**

Note: All questions are compulsory.

Q-1: (a)	What is the time complexity of the following code?
	`def fun(n,m):`
	`arr=[[0]*m for i in range(n)]`
	`for i in range(n):`
	`for j in range(m):`
	`k=1`
	`while k<n*m:`
	`k*=2`
		(2.5 marks)

(b)	What is the time complexity for the following program?
	`def fun(N):`
	`ans=0`
	`for i in range(1,N+1):`
	`for j in range(0,N,i):`
	`ans += 1`
	`print(ans)`
		(2.5 marks)

Q-2: (a)	Suppose you are given an array s[1..n] and a procedure reverse (s, i, j) which reverses the order of elements in a between positions i and j (both inclusive). What does the following sequence do, where 1 <= k <= n:
	reverse(s, 1, k) ;
	reverse(s, k + 1, n);
	reverse(s, l, n);
		(2.5 marks)

(b)	Describe deletion operation from end in single linked list with example.	**(2.5 marks)**
Q-3: (a)	Describe difference between linear and non-linear data structures? Explain with example.	**(2.5 marks)**
(b)	What is an array? How to declare one dimensional array? Explain with example.	**(2.5 marks)**

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B.Tech. II mid semester Examination, 29th May,2023.

Mid Sem-I	**Subject: Mathematics-II**	**Subject Code: MAL201**
Branch: CSE	**Max. Marks:-15**	**Time: 60 Min.**

Note: All questions are compulsory. (Rough sketch compulsory).

Q-1	Evaluate $\iiint_R \frac{dxdydz}{\sqrt{1-x^2-y^2-z^2}}$, where R is the region bounded by x=0,y=0,z=0 and $x^2+y^2+z^2=1$.	**3 Marks**
Q-2	Evaluate $\iint_A x^2dxdy$, where A is the region in the first quadrant bounded by the hyperbola xy=16 and the lines y=x ,y=0 and x=8	**3 Marks**
Q-3	Find the directional derivative of $f(x,y,z) = 4e^{x+5y-13z}$ at the point (1,2,3) in the direction towards the point (-3,5,7).	**3 Marks**
Q-4	Evaluate $\iint_S \vec{A}.\hat{n}ds$,where$\vec{A} = z\hat{\imath} + x\hat{\jmath} - 3y^2z\hat{k}$,and S is the surface of the cylinder $x^2+y^2=16$ included in the first octant between z=0 to z=5.	**3 Marks**
Q-5	For what values of b and c,$\vec{F} = (y^2+2czx)\hat{\imath} + y(bx+cz)\hat{\jmath} + (y^2+cx^2)\hat{k}$ is irrotational. Find the scalar Ø such that $\vec{F} = \nabla Ø$	**3 Marks**

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Email: sonepatiiit@gmail.com, website: www.iiitsonepat.ac.in
Department of Computer Science & Engineering (CSE)

Digital Electronics (ECL202)

Mid Sem. Exam-I (May, 2023)

Roll No. .... CSE (Semester- II) Total marks: 15

(All questions are compulsory)

Note: Solve the questions according to their marks.

1. (a) What is the minimum decimal equivalent of the number $(12B)_x$? (1)

   (b) What is the equivalent Hexadecimal representation of $(10110010)_{Gray}$? (1)

   (c) If $\overline{AB} + \overline{A}B = C$, then find $\overline{AC} + \overline{A}C$. (1)

   (d) What is the range of signed decimal numbers that can be represented by 7-bits 1's complement number? (1)

2. (a) Find the minimum number of 2-input NAND gates required to implement the following Boolean function. Draw the logic circuit diagram also.

$$F = (\bar{x} + \bar{y})(w + z)$$

(2)

   (b) Assume $X = 01100$ and $Y = 10001$ are two 5-bits binary numbers represented in two's complement format. What is the sum of X and Y represented in two's complement format using 8-bits? (2)

3. What is Multiplexer? Design a 16×1 Multiplexer using 4×1 multiplexers. (3)

4. (a) Implement the following Boolean function using 3×8 active low enable input decoder circuit.

$$F(w, y, z) = \sum_m(0, 2, 3, 4, 5, 7)$$

(2)

   (b) ICICI bank Jaipur branch vault has 3 locks with a different key for each lock. Each key is owned by different person. In order to open the door, at least two people must insert their keys into assigned locks, The signal lines A, B, and C are logic 1 if there is a key inserted into lock 1,2 or 3, respectively. What is the Boolean expression for the variable Z which is logic 1 if the door should open. Solve using K-map and draw its logic circuit diagram also. (2)

End of question paper